

MLDL EXPERIMENT 2

AIM -

Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets

DATASET SOURCE -

<https://www.kaggle.com/datasets/atharvasoundankar/viral-social-media-trends-and-engagement-analysis>

DATASET DESCRIPTION -

Viral_Social_Media_Trends.csv

This file contains **5,000 records** of viral social media posts across **TikTok, Instagram, Twitter, and YouTube**. It includes key engagement metrics such as **views, likes, shares, and comments**, along with trending **hashtags, content types**, and the **date of posting**.

Key Features:

- Covers **multiple social media platforms**
- Includes **trending hashtags and content types**
- Provides **engagement metrics** for data-driven insights
- Now includes **post date** for time-based trend analysis
- Useful for **data visualization, sentiment analysis, and trend prediction**

Column Descriptions

Column Name	Description
Post_ID	Unique identifier for each social media post
Platform	Social media platform (TikTok, Instagram, Twitter, YouTube)

Hashtag	Trending hashtag associated with the post
Content_Type	Type of content (Reel, Video, Post, Shorts, Tweet, etc.)
Region	Country where the post gained traction
Views	Total number of views received
Likes	Number of likes on the post
Shares	Number of times the post was shared
Comments	Number of comments on the post
Engagement_Level	Engagement classification: Low, Medium, or High
Post_Date	Date when the post was published (format: YYYY-MM-DD)

This dataset helps analyze **social media trends, audience engagement, and platform-specific performance** for data science & marketing research!

THEORY-

In real-world data analysis, many outcomes depend on multiple factors rather than a single variable. For example, in social media analytics, the engagement or views of a post depend on several features such as likes, shares, comments, platform type, etc.

To model such relationships, we use **Multiple Linear Regression** and its regularized versions, **Ridge Regression** and **Lasso Regression**.

In this experiment, we predict **Engagement Rate** using features such as:

- Likes
- Shares
- Comments
- Platform

These regression techniques help us understand the relationship between independent variables and the dependent variable.

- Social media platforms generate massive data.
- Businesses want to predict engagement performance.
- Simple linear regression is insufficient when multiple features are involved.
- Overfitting can occur when too many features influence the model.
- Regularization techniques (Ridge & Lasso) help improve model generalization.

Thus, we implement Multiple Regression, Ridge, and Lasso to compare their performance.

Applications

These regression techniques are widely used in:

- Social Media Analytics (Predict engagement, views)
- Marketing & Advertising (ROI prediction)
- Finance (Stock prediction)
- Healthcare (Risk prediction)
- Real Estate (House price prediction)
- Business Forecasting

In this experiment, they are applied to predict:

$$\text{Engagement Rate} = \frac{\text{Likes} + \text{Shares} + \text{Comments}}{\text{Views}}$$

Multiple Linear Regression

Multiple Linear Regression models the relationship between one dependent variable and two or more independent variables.

- To predict continuous outcomes
- To understand feature influence
- To model linear relationships

Mathematical Formula

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_n X_n$$

Where:

- Y = Dependent variable (Engagement Rate)
- β_0 = Intercept
- $\beta_1, \beta_2, \beta_3$ = Coefficients
- X_1, X_2, X_3 = Independent variables (Likes, Shares, Comments, Platform)

Objective Function

$$\text{Minimize : } \sum (y - \hat{y})^2$$

It minimizes the **sum of squared errors**.

Ridge Regression (L2 Regularization)

Definition

Ridge Regression is an extension of Linear Regression that adds a penalty term to reduce large coefficients.

- Prevents overfitting

- Handles multicollinearity
- Controls model complexity
- Improves generalization

Mathematical Formula

$$\text{Minimize : } \sum (y - \hat{y})^2 + \lambda \sum \beta^2$$

Where:

- λ = Regularization parameter
- β^2 = Square of coefficients
- Penalizes large coefficients
- Shrinks coefficients towards zero
- Does NOT make them exactly zero

Lasso Regression (L1 Regularization)

Lasso Regression adds an absolute value penalty term to the regression model.

- Performs feature selection
- Reduces model complexity
- Eliminates irrelevant features

Mathematical Formula

$$\text{Minimize : } \sum (y - \hat{y})^2 + \lambda \sum |\beta|$$

Where:

$|\beta|$ = Absolute value of coefficients

- Can shrink some coefficients exactly to 0

- Removes less important features automatically

Difference Between the Three

Model	Regularization	Effect
Linear Regression	None	May overfit
Ridge	L2 (β^2)	Shrinks coefficients
Lasso	L1 (β)	β

Why Regularization Is Important

In real-world datasets:

- Data may contain noise
- Features may be correlated
- Model may memorize training data

Regularization:

- Reduces overfitting
- Improves prediction on new data
- Controls model complexity
- Makes model more stable

OBSERVATIONS -

CODE :

```
#
=====
=====
# MULTIPLE REGRESSION + RIDGE + LASSO
# Engagement Rate Prediction System
#
=====
=====

# Step 1: Install (if needed)
# (Uncomment if running first time)
# !pip install scikit-learn pandas numpy

# Step 2: Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import
train_test_split
from sklearn.linear_model import
LinearRegression, Ridge, Lasso
from sklearn.preprocessing import
StandardScaler
from sklearn.metrics import r2_score,
mean_squared_error

# Step 3: Upload Dataset in Colab
from google.colab import files
uploaded = files.upload()

# Replace filename if different
file_name = list(uploaded.keys())[0]

# Load dataset
df = pd.read_csv(file_name)

print("Dataset Loaded Successfully )
print(df.head())

#
=====
=====
# Step 4: Create Engagement Rate Column
```

```
#
=====
=====
df["Engagement_Rate"] = (df["Likes"] +
df["Shares"] + df["Comments"]) / df["Views"]

#
=====
=====
# Step 5: Encode Platform (One Hot Encoding)
#
=====
=====
df = pd.get_dummies(df, columns=["Platform"],
drop_first=True)

#
=====
=====
# Step 6: Define Features & Target
#
=====
=====
feature_columns = ["Likes", "Shares",
"Comments"] + \
[col for col in df.columns if
col.startswith("Platform_")]

X = df[feature_columns]
y = df["Engagement_Rate"]

#
=====
=====
# Step 7: Train-Test Split
#
=====
=====
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
```

```

#
=====
=====
# Step 8: Feature Scaling
#
=====
=====

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

#
=====
=====
# Step 9: Train Models
#
=====
=====

lr = LinearRegression()
ridge = Ridge(alpha=1.0)
lasso = Lasso(alpha=0.001)

lr.fit(X_train_scaled, y_train)
ridge.fit(X_train_scaled, y_train)
lasso.fit(X_train_scaled, y_train)

#
=====
=====
# Step 10: Evaluate Models
#
=====
=====

print("\nModel Performance 📊")
print("Linear Regression R2:", r2_score(y_test,
lr.predict(X_test_scaled)))
print("Ridge Regression R2:", r2_score(y_test,
ridge.predict(X_test_scaled)))
print("Lasso Regression R2:", r2_score(y_test,
lasso.predict(X_test_scaled)))

#
=====
=====
# Step 11: User Input Section

```

```

#
=====
=====

print("\nEnter Post Details to Predict
Engagement:\n")
likes = float(input("Likes: "))
shares = float(input("Shares: "))
comments = float(input("Comments: "))
views = float(input("Views: "))
platform = input("Platform
(TikTok/Instagram/Twitter/YouTube etc.): ")

# Calculate actual engagement rate from input
actual_eng_rate = (likes + shares + comments) /
views

# Prepare input dictionary
input_dict = {
    "Likes": likes,
    "Shares": shares,
    "Comments": comments
}

for col in feature_columns:
    if col.startswith("Platform_"):
        input_dict[col] = 1 if col ==
f"Platform_{platform}" else 0

input_df = pd.DataFrame([input_dict])

# Scale input
input_scaled = scaler.transform(input_df)

# Predictions
lr_pred = lr.predict(input_scaled)[0]
ridge_pred = ridge.predict(input_scaled)[0]
lasso_pred = lasso.predict(input_scaled)[0]

#
=====
=====
# Step 12: Engagement Classification
#
=====
=====

def classify(score):
    if score < 0.05:

```

```

        return "Low Engagement"
    elif score < 0.15:
        return "Medium Engagement"
    else:
        return "High Engagement"

print("\n===== RESULT
=====")
print(f"\nActual Engagement Rate (calculated):
{actual_eng_rate:.4f}")

print("\nPredicted Engagement Rate:")
print(f"Linear Regression: {lr_pred:.4f}")
print(f"Ridge Regression: {ridge_pred:.4f}")
print(f"Lasso Regression: {lasso_pred:.4f}")

print("\nPredicted Engagement Level (Using
Ridge):")
print(classify(ridge_pred))

#
=====
=====
# Step 13: Model Explanation
#
=====
=====

```

```

print("\n----- MODEL EXPLANATION
-----")

print("\n1 Linear Regression:")
print("Minimizes Squared Error:")
print("  $\sum (y - \hat{y})^2$ ")

print("\n2 Ridge Regression (L2
Regularization):")
print(" Minimize:  $\sum (y - \hat{y})^2 + \lambda \sum (\beta^2)$ ")
print(" → Reduces large coefficients")
print(" → Prevents overfitting")

print("\n3 Lasso Regression (L1
Regularization):")
print(" Minimize:  $\sum (y - \hat{y})^2 + \lambda \sum |\beta|$ ")
print(" → Can shrink some coefficients to zero")
print(" → Performs feature selection")

print("\nRegularization helps reduce model
complexity and improve generalization.")

print("\n=====
=====")

```

OUTPUT :

```

... Choose files Cleaned_Vi..._Trends.csv
Cleaned_Viral_Social_Media_Trends.csv(text/csv) - 410182 bytes, last modified: 12/02/2026 - 100% done
Saving Cleaned_Viral_Social_Media_Trends.csv to Cleaned_Viral_Social_Media_Trends.csv
Dataset Loaded Successfully [X]

Post_ID  Post_Date  Platform  Hashtag  Content_Type  Region  Views  \
0  Post_1  2022-01-13  TikTok  #Challenge  Video  UK  4163464
1  Post_2  2022-05-13  Instagram  #Education  Shorts  India  4155940
2  Post_3  2022-01-07  Twitter  #Challenge  Video  Brazil  3666211
3  Post_4  2022-12-05  YouTube  #Education  Shorts  Australia  917951
4  Post_5  2023-03-23  TikTok  #Dance  Post  Brazil  64866

Likes  Shares  Comments  Engagement_Level
0  339431  53135  19346  High
1  215240  65860  27239  Medium
2  327143  39423  36223  Medium
3  127125  11687  36806  Low
4  171361  69581  6376  Medium

Model Performance [M]
Linear Regression R2: -0.0011483407444909943
Ridge Regression R2: -0.00114831136998772
Lasso Regression R2: -0.001139469605947241

```

```
Enter Post Details to Predict Engagement:

Likes: 339431
Shares: 53135
Comments: 19346
Views: 4163464
Platform (TikTok/Instagram/Twitter/YouTube etc.): TikTok

===== RESULT =====

Actual Engagement Rate (calculated): 0.0989

Predicted Engagement Rate:
Linear Regression: 0.7241
Ridge Regression: 0.7240
Lasso Regression: 0.7187

Predicted Engagement Level (Using Ridge):
High Engagement

----- MODEL EXPLANATION -----
```

CONCLUSION -

In this experiment, Multiple Linear Regression was implemented to predict **quantity_sold** using independent variables such as **price**, **discount_percent**, **rating**, and **review_count**.

The model was trained on the dataset and evaluated using **R² Score** and **Mean Squared Error (MSE)** to measure prediction accuracy.

From the results:

- The regression model successfully established a linear relationship between the dependent and independent variables.
- The coefficient values helped identify which features had the most influence on quantity sold.
- The Actual vs Predicted graph showed how closely the model's predictions matched real values.
- The Residual plot helped verify whether errors were randomly distributed.
- **Ridge Regression (L2 Regularization)** reduces the magnitude of coefficients by adding a squared penalty term. It helps handle multicollinearity and stabilizes the model.
- **Lasso Regression (L1 Regularization)** adds an absolute penalty term, which can shrink some coefficients to zero and perform feature selection.